

ENTERPRISE IT LAB

Windows Server Hybrid Infrastructure & Zendesk ITSM Simulation

Microsoft Azure • Windows Server 2022 • Active Directory Domain Services •
Zendesk

Prepared by: Caleb Cathrine
Domain: CalebC.local | DC: DC-01 | Client: Client-01
Date: May 2026

Project Overview

This portfolio documents the end-to-end design, deployment, and administration of a corporate IT infrastructure alongside an enterprise IT Service Management (ITSM) ticketing workflow. Using Microsoft Azure, a Windows Server 2022 Domain Controller (DC-01) was provisioned running Active Directory Domain Services (AD DS), alongside a Windows 10 Enterprise endpoint (Client-01). Zendesk was integrated to simulate enterprise technical support covering tier-1 and tier-2 user lifecycle, access control management, and security requests.

Cloud Platform	Microsoft Azure (Free Tier)
Server OS	Windows Server 2022 Datacenter
Client OS	Windows 10 Enterprise
Directory Service	Active Directory Domain Services (AD DS)
Domain Name	CalebC.local
DC Hostname	DC-01 Static IP: 10.0.0.4
Client Hostname	Client-01
ITSM Platform	Zendesk (14-day free trial)
Agent	Caleb Cathrine / Support/Caleb Cathrine
End Users	Jane Doe (jane_admin) John Doe (jdoe) HR Department

Core Competencies Demonstrated

- Cloud Infrastructure & Networking: Virtual network design, custom DNS routing, subnet allocations, and cloud NIC troubleshooting in Microsoft Azure.
- Systems Administration: Windows Server 2022 deployment, AD DS installation, domain promotion, and domain join implementations.
- Identity & Access Management (IAM): Lifecycle management (provisioning/deprovisioning), RBAC via AD Security Groups, and password security policies.
- Network Storage Administration: Enterprise SMB file-sharing, NTFS network permissions, and client-side network drive mapping (Z: drive).
- Enterprise ITSM & Service Desk Operations: Professional ticket scoping, classification, internal documentation, SLAs, and customer-facing communication using Zendesk.

PHASE 1

Virtual Cloud Datacenter Creation

Azure Virtual Network, Subnet, and VM Provisioning

To establish an isolated corporate network, an enterprise-grade tenant environment was deployed using Microsoft Azure. A custom Virtual Network (DC-01-vnet) was configured with a dedicated subnet block (10.0.0.0/24). DC-01 was provisioned as a Windows Server 2022 Datacenter virtual machine instance on a Standard B2s v2 SKU.

The screenshot shows the 'Networking' tab in the Azure portal for creating a virtual machine. The configuration is as follows:

- Virtual network:** (new) DC-01-vnet (with a 'Create new' link below it)
- Subnet:** (new) default (10.0.0.0/24)
- Public IP:** (new) DC-01-ip (with a 'Create new' link below it and a note: 'Public IP addresses have a nominal charge. [Estimate price](#)')
- NIC network security group:** Basic (selected), with options for None and Advanced.
- Public inbound ports:** Allow selected ports (selected), with an option for None.

Figure: Azure VM Networking tab — DC-01-vnet virtual network with default subnet 10.0.0.0/24, DC-01-ip public IP, Basic NIC NSG, and inbound ports allowed.

The screenshot shows the 'Review + create' step of the Azure VM provisioning process. The configuration is confirmed as follows:

- Price:** 1 X Standard B2s v2 by Microsoft. Subscription credits apply. **0.1180 USD/hr**. Pricing for other VM sizes.
- TERMS:** A text block stating that by clicking 'Create', the user agrees to the legal terms and privacy statement(s) associated with the Marketplace offering(s).

At the bottom, there are navigation buttons: '< Previous', 'Next >', and a blue 'Create' button. A link 'Download a template for automation' is also visible.

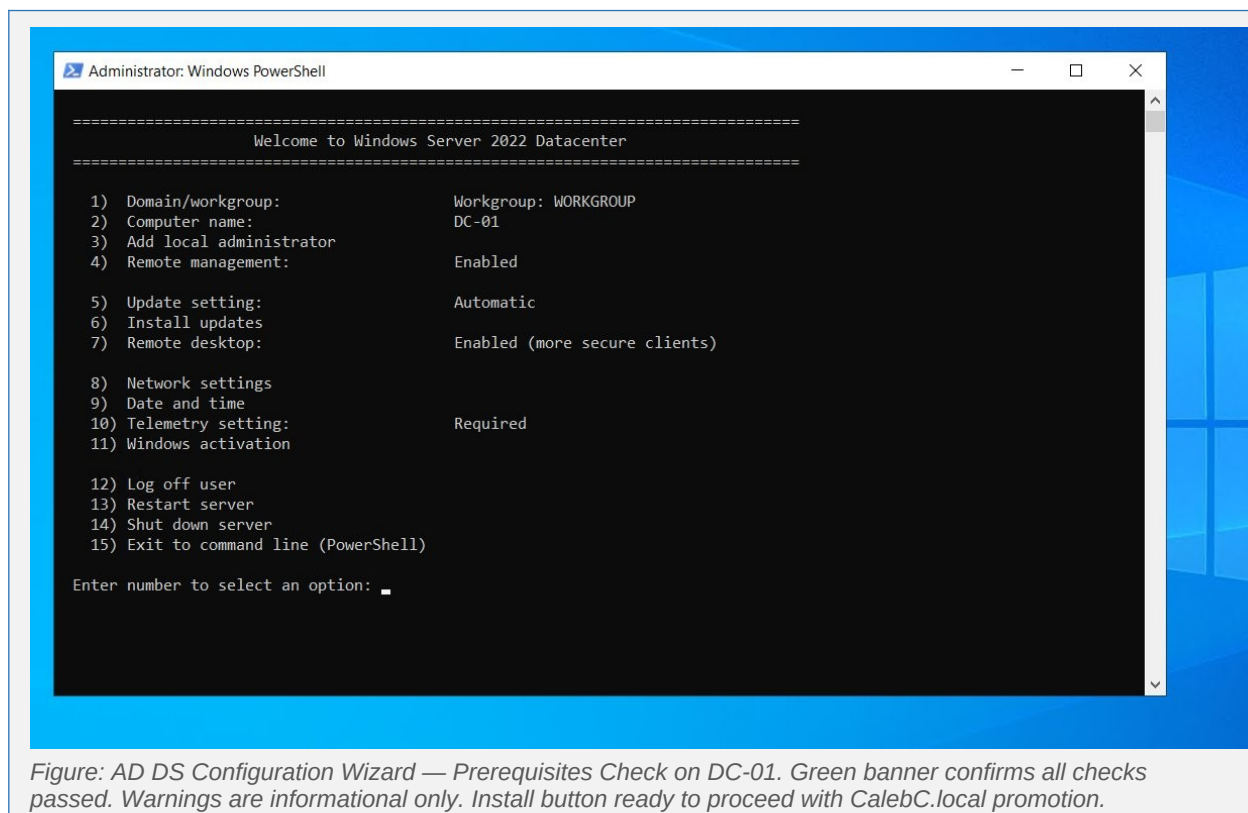
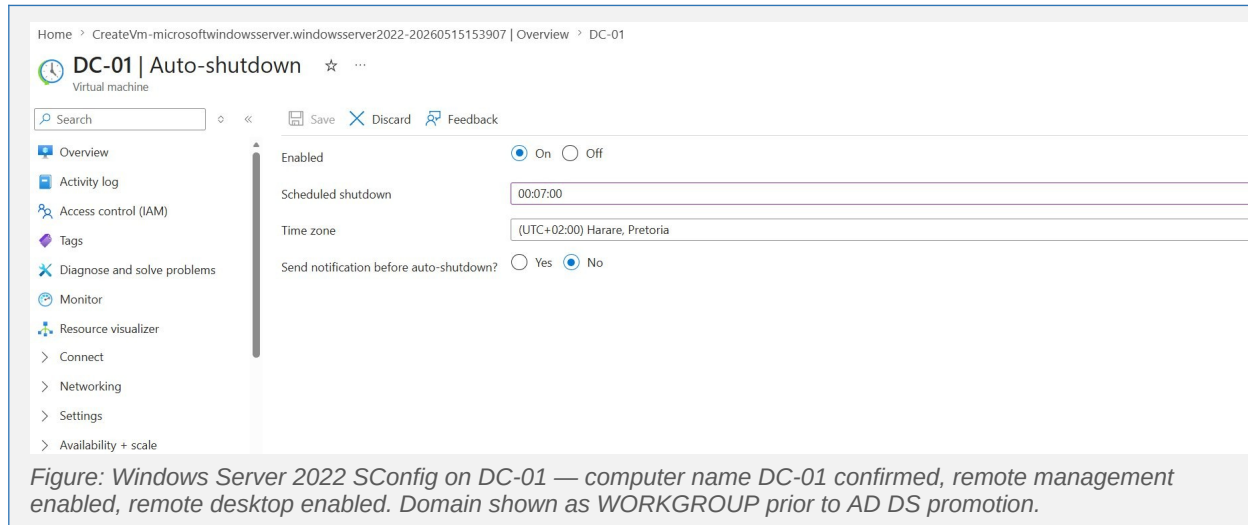
Figure: Azure VM Review + Create — Validation passed for DC-01 (Standard B2s v2, 0.1180 USD/hr). Final confirmation screen before provisioning.

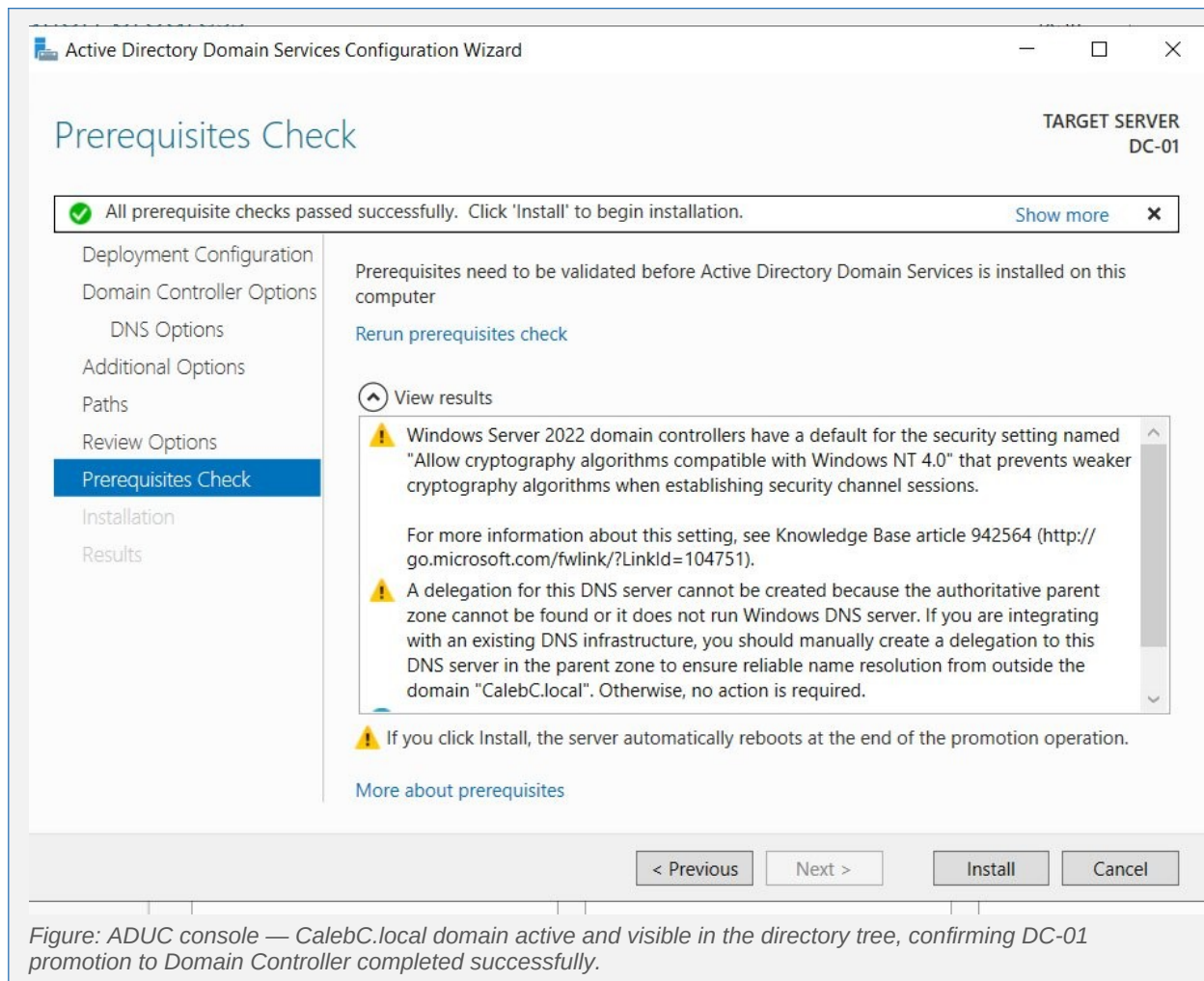
PHASE 2

Active Directory Core Provisioning

AD DS Role Installation and Domain Controller Promotion

Once the host server instance was stable, it was converted into a dedicated Domain Controller for the CalebC.local corporate namespace. The AD DS binaries were installed via Server Manager and all prerequisite checks were cleared before promoting DC-01 to the primary Domain Controller.





PHASE 3

Client Subnet Configuration & Domain Enrollment

DNS Override, Domain Join, and Endpoint Validation

Before joining Client-01 to the domain, network layer requirements were manually configured. The Client-01 NIC private IP DNS was pointed to DC-01's static internal IP (10.0.0.4). The workstation was then enrolled into CalebC.local using domain administrator credentials.

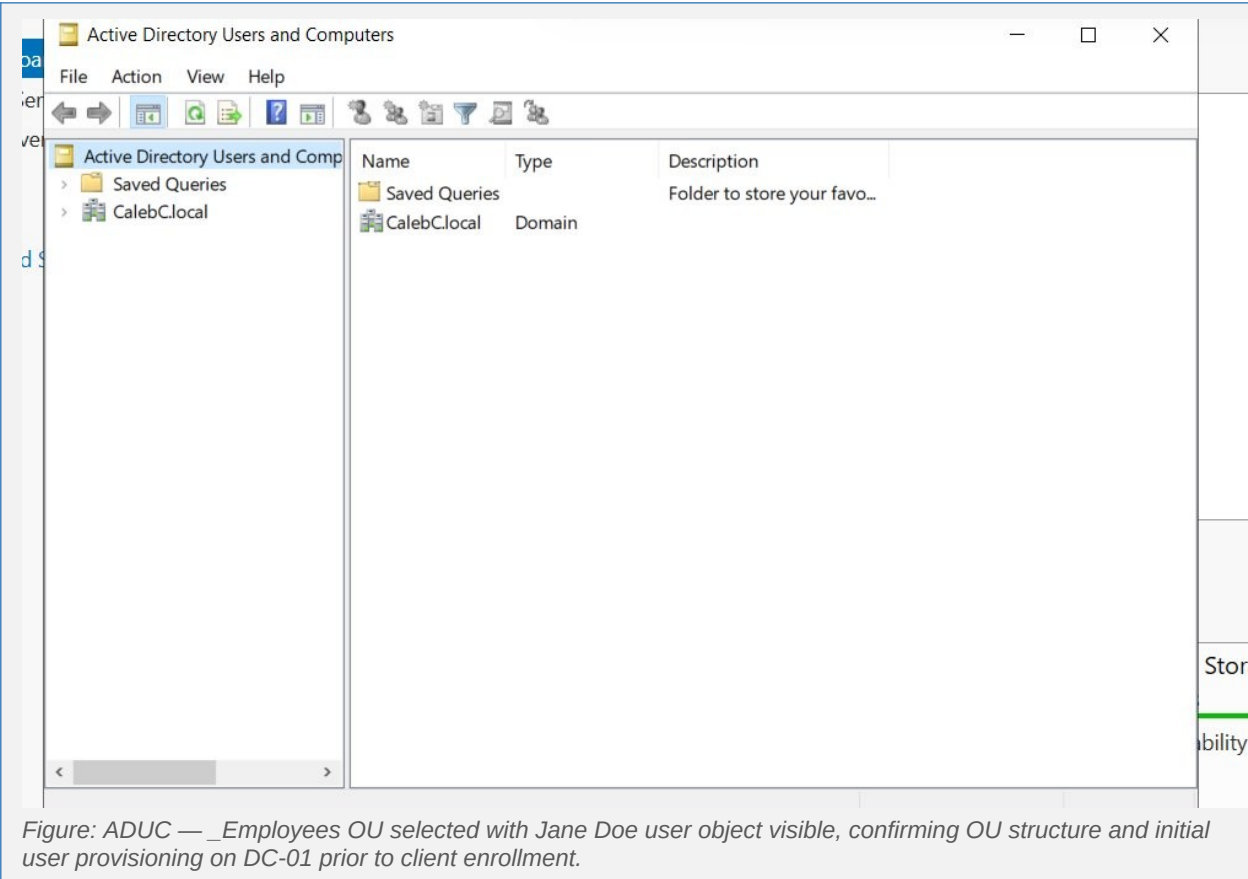


Figure: ADUC — _Employees OU selected with Jane Doe user object visible, confirming OU structure and initial user provisioning on DC-01 prior to client enrollment.

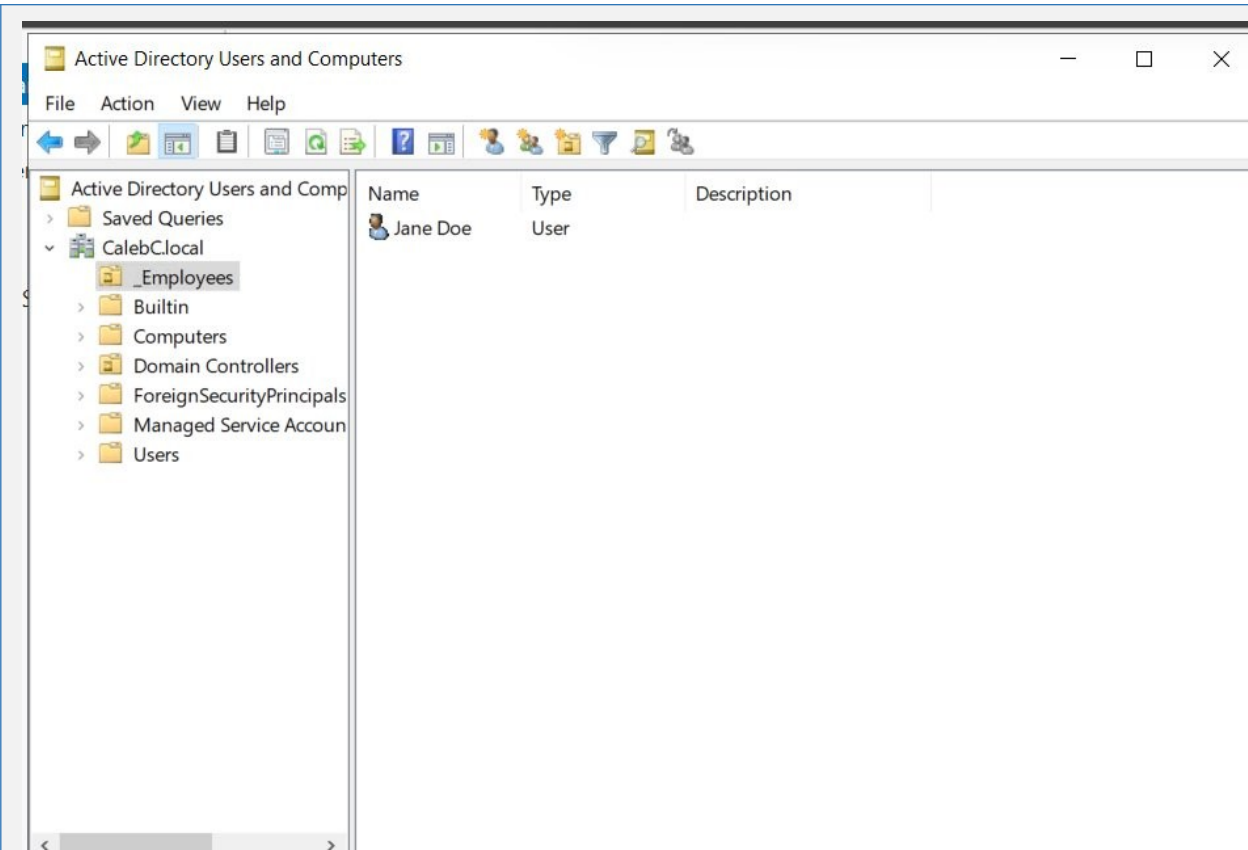


Figure: Azure Marketplace — Windows 10 Enterprise image options reviewed during Client-01 provisioning, showing available enterprise endpoint OS versions for Azure VM deployment.

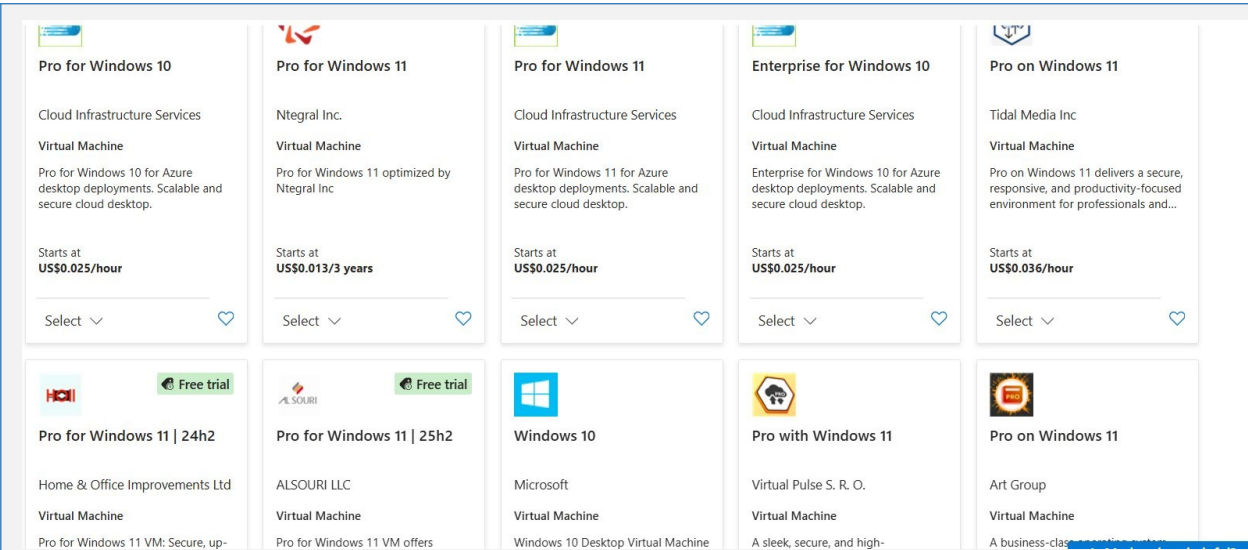


Figure: Azure Marketplace — additional Windows VM image options shown during Client-01 setup, confirming enterprise image selection process.

Windows 10 Enterprise multi-session, version 22H2 - x64 Gen 1

Windows 10 Enterprise multi-session, version 22H2 - x64 Gen 2

Windows 10 Enterprise, version 22H2 - x64 Gen 1

Windows 10 Enterprise, version 22H2 - x64 Gen 2

Windows 10 Enterprise N, version 22H2 - x64 Gen 1

Windows 10 Enterprise N, version 22H2 - x64 Gen 2

Windows 10 Enterprise LTSC 2021 - x64 Gen 1

Windows 10 Enterprise LTSC 2021 - x64 Gen 2

Windows 10 Enterprise N LTSC 2021 - x64 Gen 1

Windows 10 Enterprise N LTSC 2021 - x64 Gen 2

Windows 10 Enterprise 2019 LTSC - x64 Gen 1

Windows 10 Enterprise 2019 LTSC - x64 Gen 2

Windows 10 Enterprise N 2019 LTSC - x64 Gen 1

Windows 10 Enterprise N 2019 LTSC - x64 Gen 2

Windows 10 Enterprise 2016 LTSC - x64 Gen 1

Windows 10 Enterprise 2016 LTSC - x64 Gen 2

Figure: Computer Name/Domain Changes on Client-01 — Domain field shows CalebC.local with the "Welcome to the CalebC.local domain" confirmation popup, confirming successful domain join.

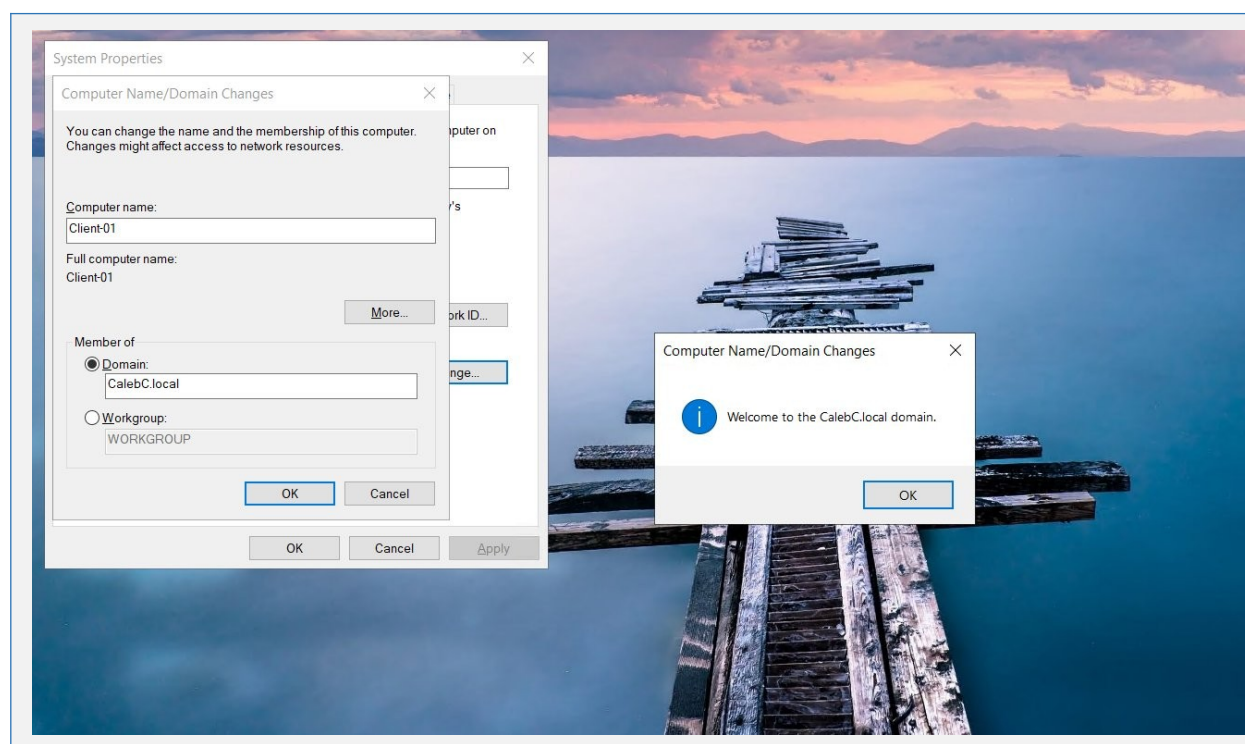


Figure: ADUC on DC-01 — _Employees OU showing Jane Doe user object confirmed present post domain join, verifying identity visibility across the domain.

CASE STUDY 1

Enterprise Share Mapping & SMB File Services

Zendesk Ticket #2 — Cannot Access Shared Network Drive

Business Requirement

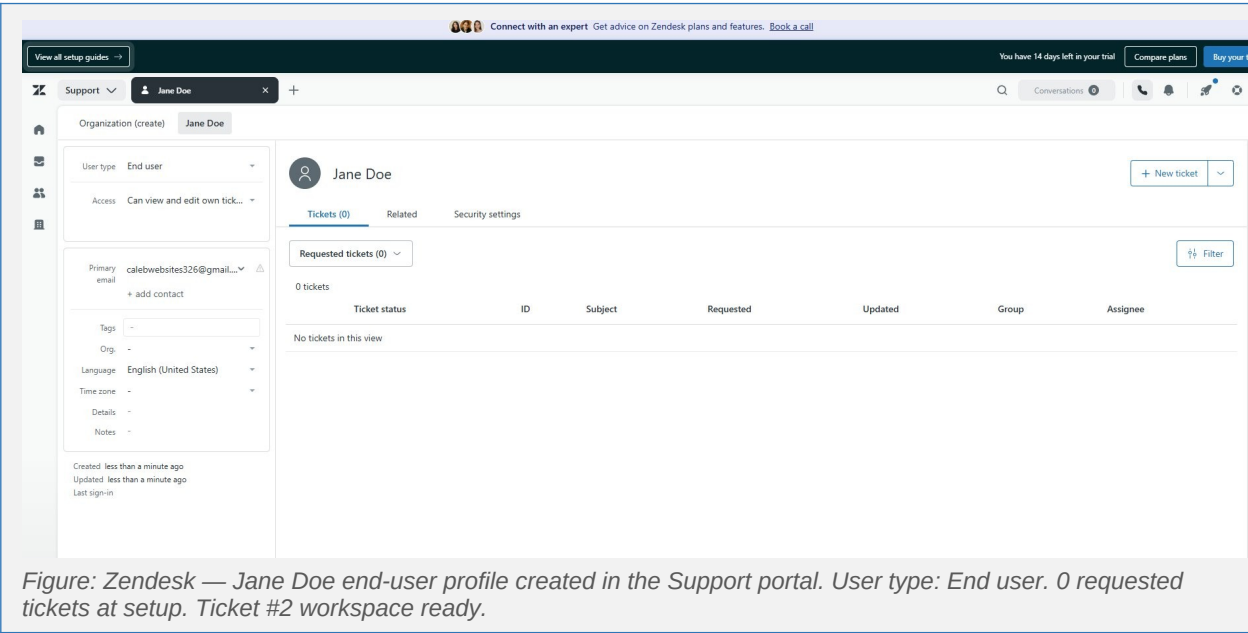
The business required a centralized file server mapping directory to allow authenticated end-users to exchange documents securely. This task was tracked through Zendesk Ticket #2, submitted by user Jane Doe, reporting she could not discover or view company-wide file folders on her newly provisioned Client-01 machine.

ZENDESK TICKET

From: Jane Doe
Subject: Cannot access shared network drive
Body: Hi Help Desk, I just logged into my new Client-01 machine as jane_admin but I can't see the public folder. Please help!

Execution Strategy & Implementation

Logged into DC-01 and accessed ADUC. An _Employees OU was created to catalog standard corporate user entries. Jane Doe's user identity object was built inside the _Employees container (UPN: jane_admin). Standard SMB sharing policies were enabled on the CompanyShare folder for Authenticated Users. Remote access into Client-01 confirmed the directory mapped successfully as the Z: drive.



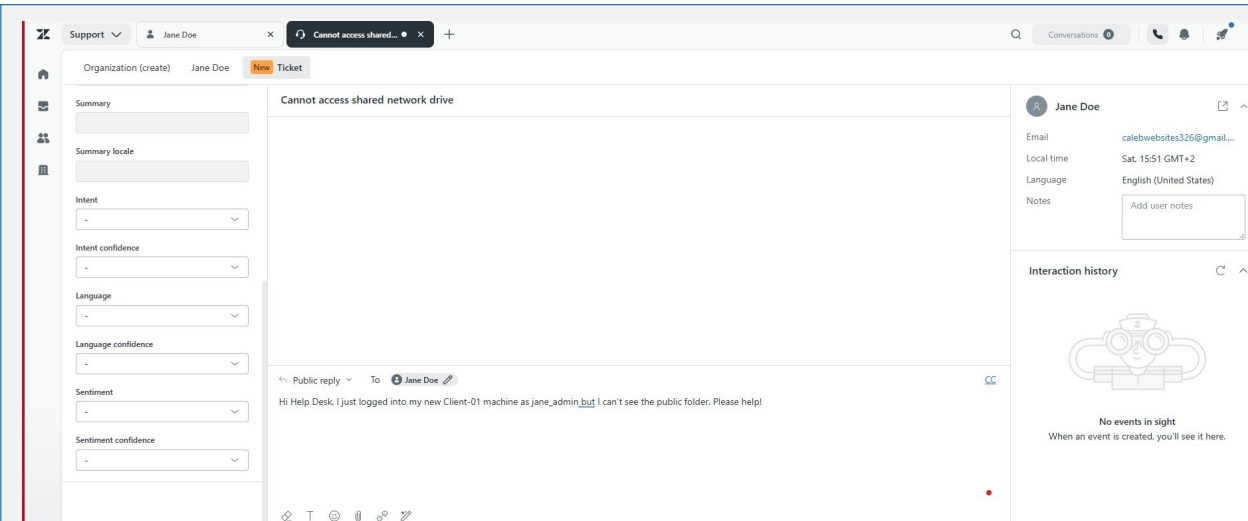


Figure: CompanyShare Properties — Sharing tab on DC-01 confirming network path \\DC-01\\CompanyShare is active. Advanced Sharing enabled for NTFS permission configuration.

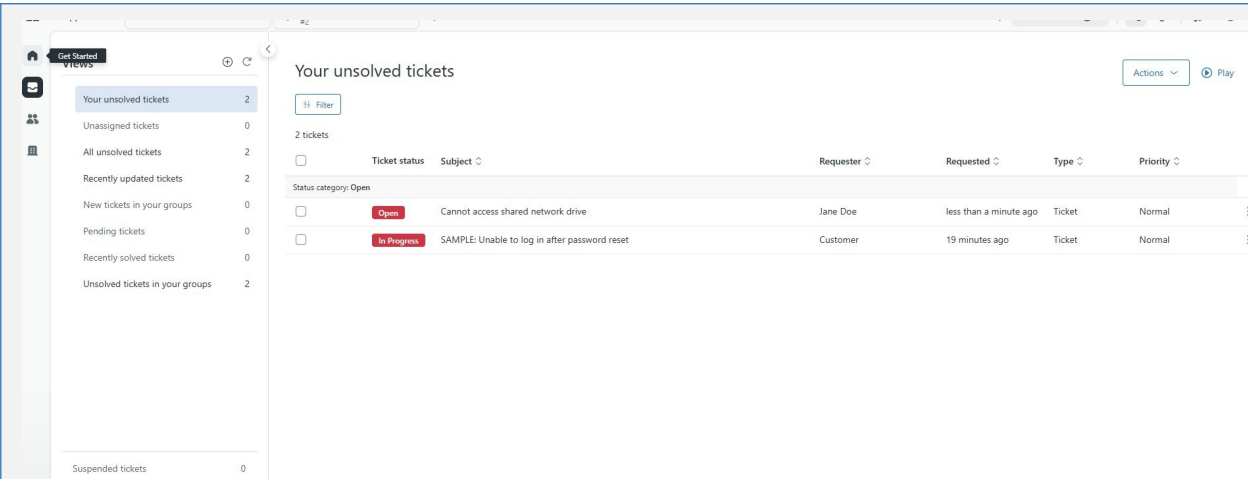


Figure: Client-01 File Explorer — This PC view showing CompanyShare (\\DC-01\\Z:) mounted under Network Locations with 112 GB free. Z: drive mapping confirmed active.

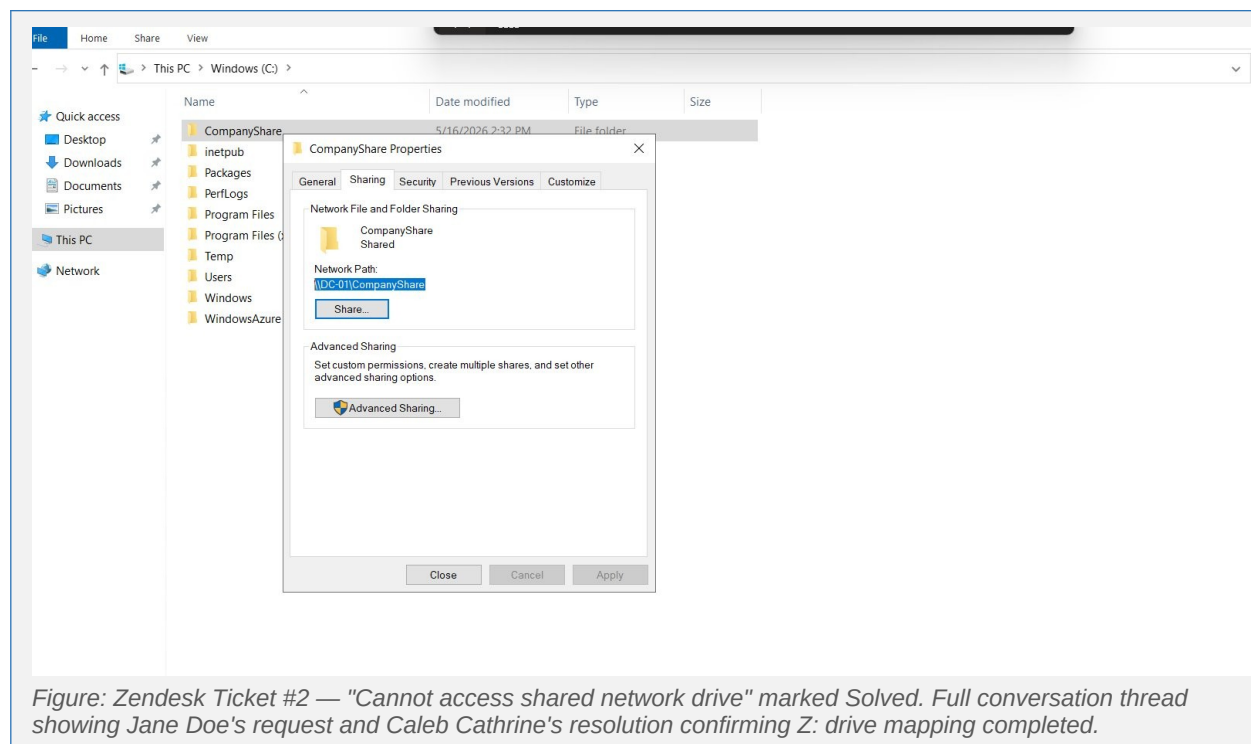


Figure: Zendesk Ticket #2 — "Cannot access shared network drive" marked Solved. Full conversation thread showing Jane Doe's request and Caleb Cathrine's resolution confirming Z: drive mapping completed.

CASE STUDY 2

IAM Operations — Lockouts & Emergency Password Resets

Zendesk Ticket #3 — Urgent: Locked Out / Password Reset

Business Requirement

User accounts routinely trigger policy-driven lockouts due to repeated login failures. This was handled via Zendesk Ticket #3 as an urgent priority response, where Jane Doe reported her credentials had locked her out of her machine session entirely.

ZENDESK TICKET

From: Jane Doe

Subject: Urgent: Locked out of my computer / Password reset

Body: *Hi Help Desk, I typed my password in wrong too many times this morning and now it says my account is locked out. Can you please reset my password so I can get to work? Thanks!*

Execution Strategy & Implementation

ADUC was opened to locate Jane Doe's identity object in the _Employees OU. The Reset Password dialog was used to set a temporary password, unlock the account, and enforce password change at next logon. The ticket was resolved in Zendesk with temporary credentials communicated to the user.

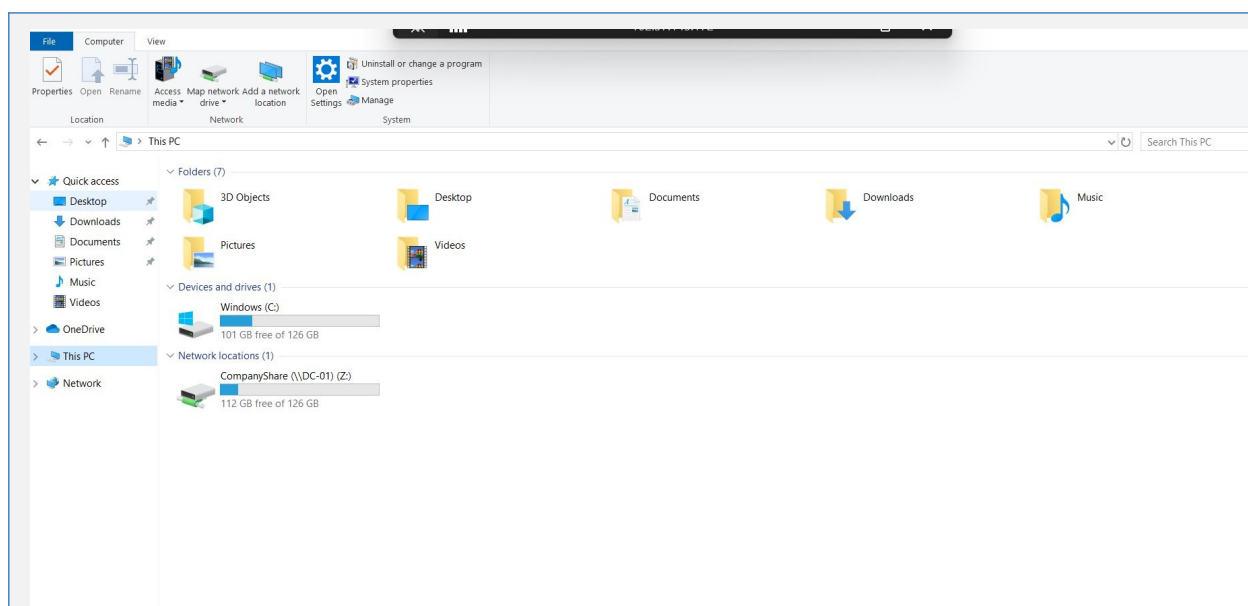
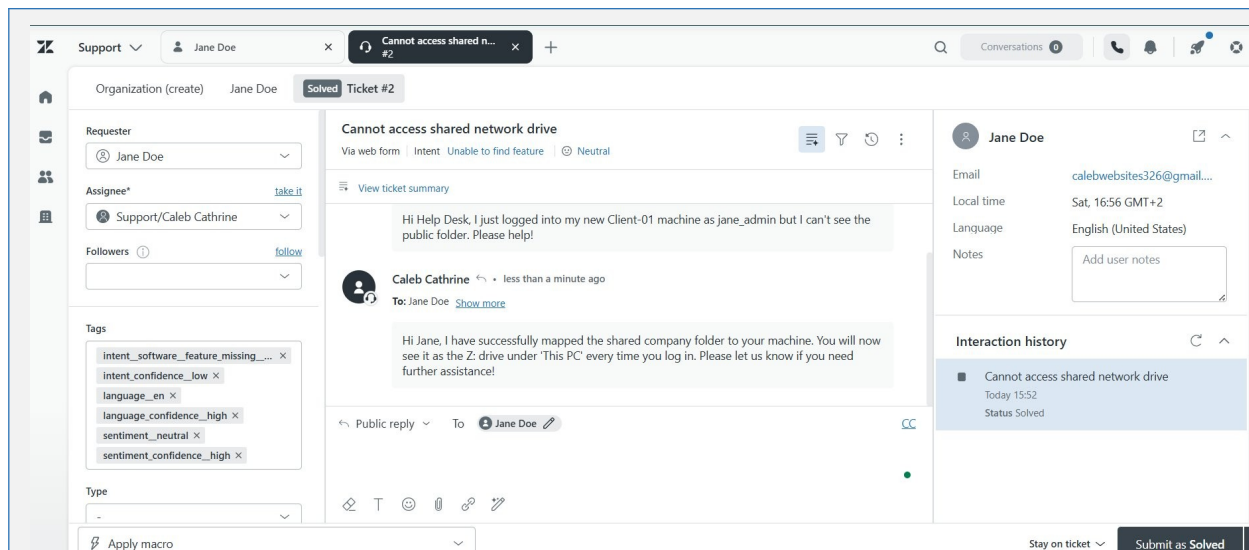
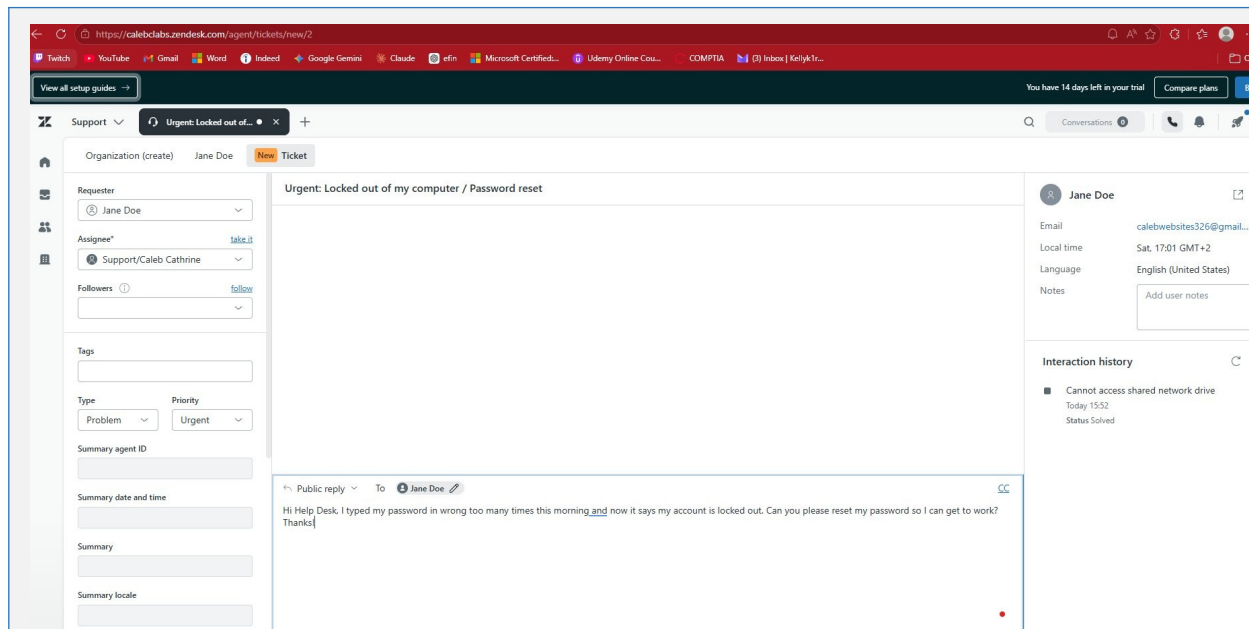


Figure: Zendesk Ticket #3 — "Urgent: Locked out of my computer / Password reset" open. Requester: Jane Doe. Type: Problem. Priority: Urgent. Assigned to Support/Caleb Cathrine. Jane Doe's lockout message visible.



The screenshot shows the Zendesk interface for Ticket #2, titled "Cannot access shared network drive". The ticket is marked as "Solved". The requester is Jane Doe, and the assignee is Support/Caleb Cathrine. The ticket summary states: "Via web form | Intent: Unable to find feature | Sentiment: Neutral". The conversation history shows a message from Jane Doe: "Hi Help Desk, I just logged into my new Client-01 machine as jane_admin but I can't see the public folder. Please help!". A response from Caleb Cathrine follows: "Hi Jane, I have successfully mapped the shared company folder to your machine. You will now see it as the Z: drive under 'This PC' every time you log in. Please let us know if you need further assistance!". The ticket is currently in a "Public reply" state, addressed to Jane Doe. The right sidebar shows the requester's profile (Jane Doe, email: calebwesites326@gmail.com, local time: Sat, 16:56 GMT+2, language: English (United States)) and the interaction history, which includes the ticket title and the time it was solved (Today 15:52).

Figure: ADUC — Reset Password dialog for Jane Doe in _Employees OU. New password set, "User must change password at next login" checked, "Unlock the user's account" checked. Status shown as Unlocked.



The screenshot shows the Zendesk interface for Ticket #3, titled "Urgent: Locked out of my computer / Password reset". The ticket is marked as "New". The requester is Jane Doe, and the assignee is Support/Caleb Cathrine. The ticket summary states: "Urgent: Locked out of my computer / Password reset". The conversation history shows a message from Jane Doe: "Hi Help Desk, I typed my password in wrong too many times this morning and now it says my account is locked out. Can you please reset my password so I can get to work? Thanks!". The ticket is currently in a "Public reply" state, addressed to Jane Doe. The right sidebar shows the requester's profile (Jane Doe, email: calebwesites326@gmail.com, local time: Sat, 17:01 GMT+2, language: English (United States)) and the interaction history, which includes the ticket title and the time it was solved (Today 15:52).

Figure: Zendesk Ticket #3 — marked Solved (Problem #3). Full thread showing Jane Doe's lockout request and resolution reply with temporary password Welcome2026! communicated.

CASE STUDY 3

HR System Lifecycle — Employee Offboarding Operations

Zendesk Ticket #4 — Employee Offboarding Request: Jane Doe

Business Requirement

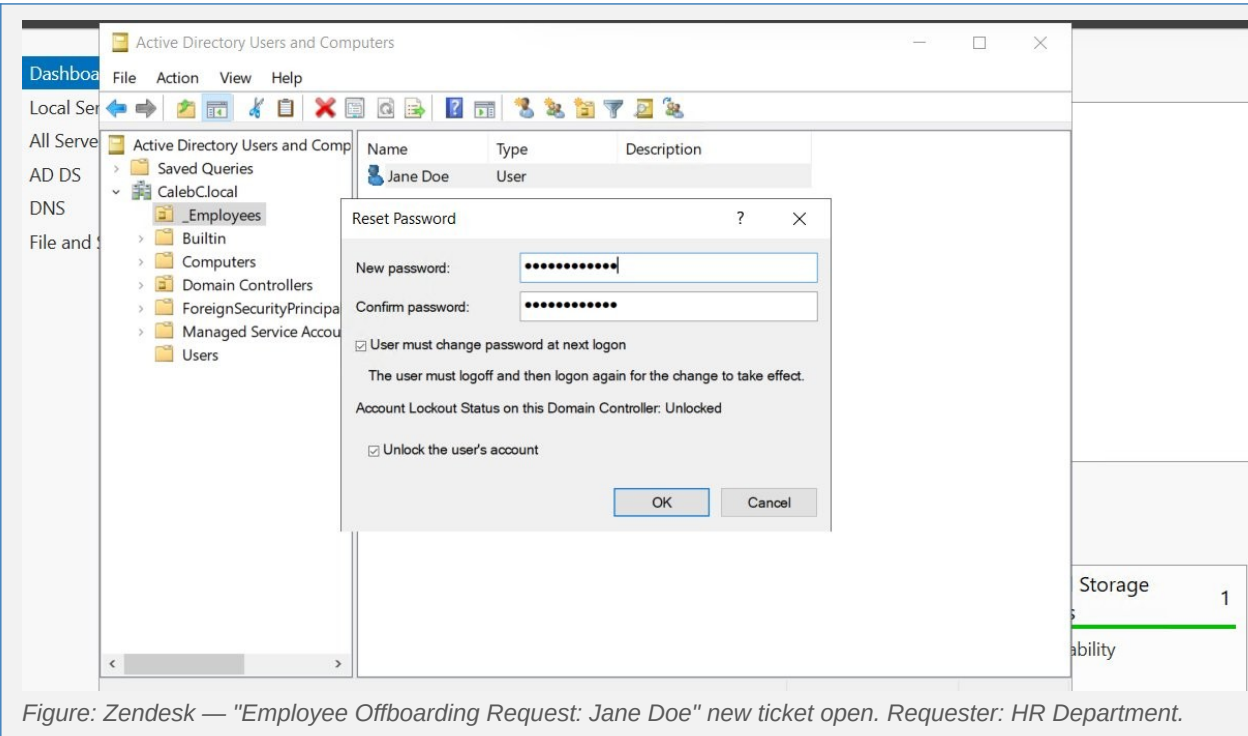
When personnel departures occur, HR triggers identity governance procedures requiring IT to de-provision network access. This arrived via Zendesk Ticket #4 from the HR Department, ordering the immediate revocation of Jane Doe's identity privileges.

ZENDESK TICKET

From: HR Department
Subject: Employee Offboarding Request: Jane Doe
Body: *Hi IT, today is Jane Doe's last day with the company. Please disable her Active Directory user account and remove her from all network access by the end of the day. Thank you, HR.*

Execution Strategy & Implementation

DC-01 was accessed and ADUC opened to target Jane Doe's user record in the _Employees OU. The Disable Account control was executed. The systemic down-arrow icon confirmed all privileges were frozen. The ticket was closed in Zendesk with documented confirmation of the offboarding action.



Assigned to Support/Caleb Cathrine. HR's message requesting immediate account disable visible.

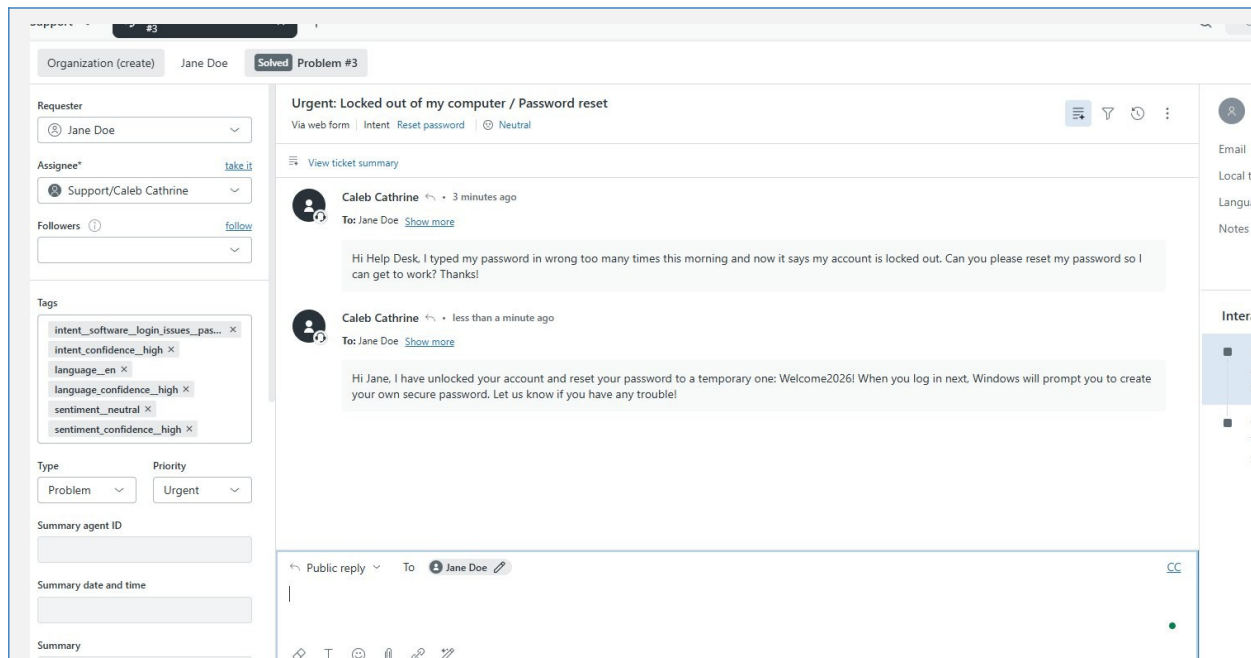


Figure: ADUC — Jane Doe user object in _Employees OU showing the disabled account icon (downward arrow overlay), confirming account successfully de-provisioned as part of the offboarding process.

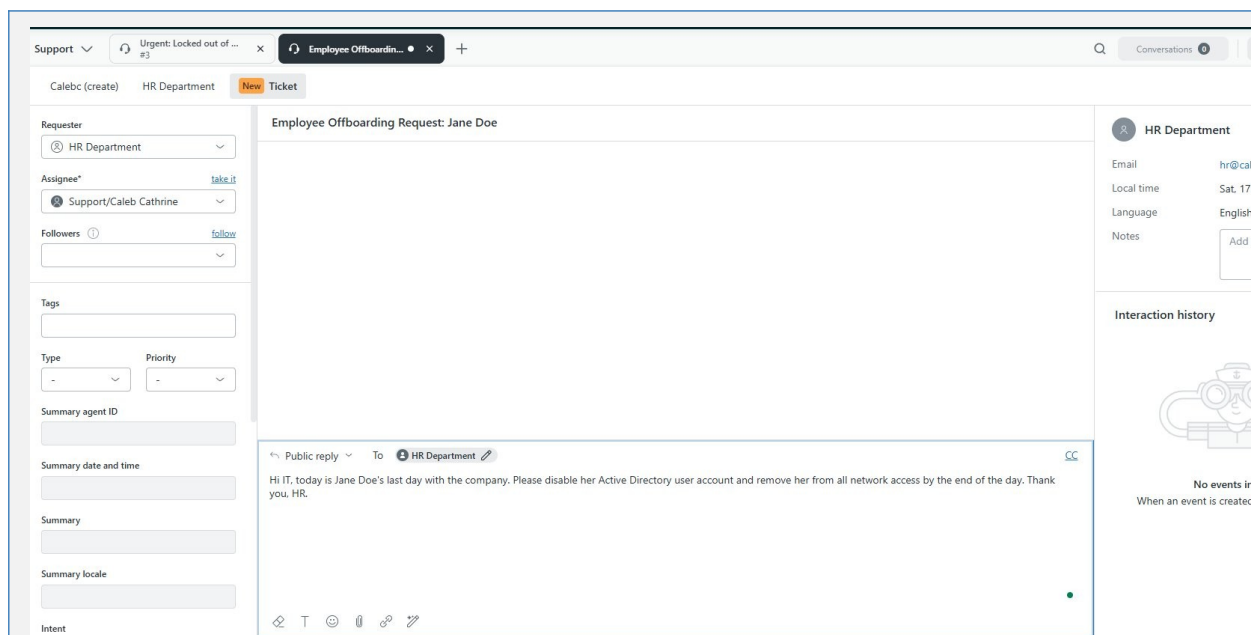


Figure: Zendesk — Solved ticket view showing interaction history with both prior tickets (Urgent lockout: Solved, Cannot access shared drive: Solved) visible, demonstrating full lifecycle documentation.

CASE STUDY 4

Advanced RBAC Permissions & Enterprise Directory Nesting

Zendesk Ticket #5 — Access Request: Add John Doe to Sales Groups

Business Requirement

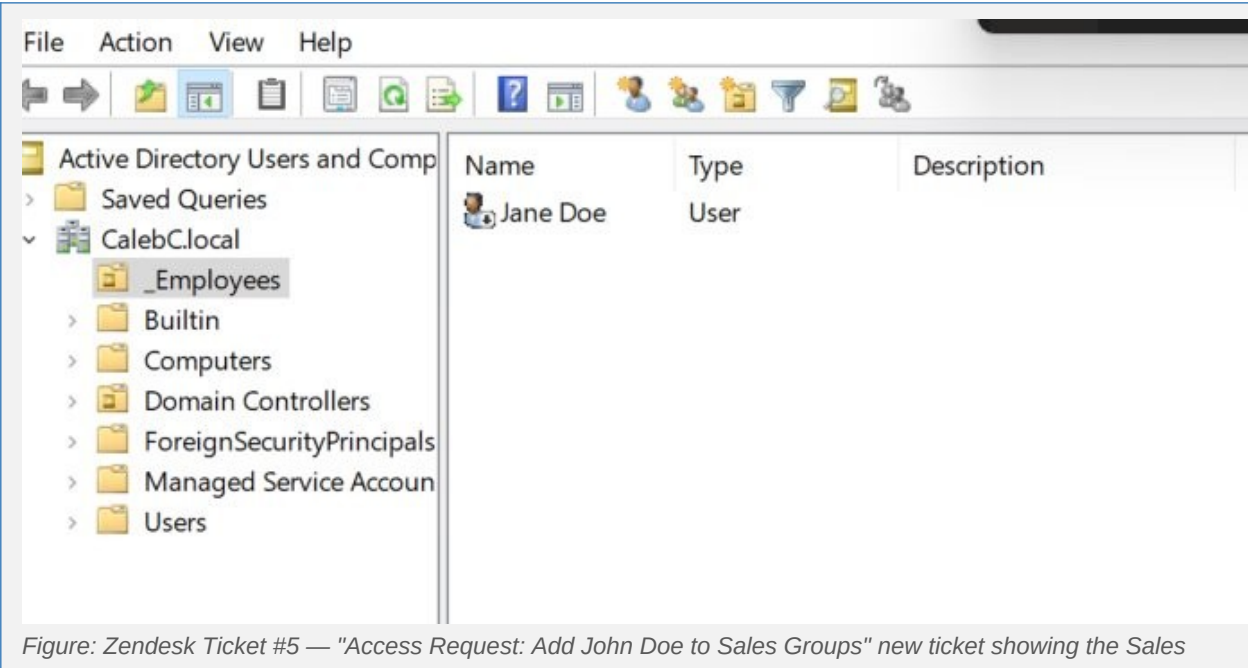
To avoid manually applying file permissions user-by-user, scalable IT environments use Role-Based Access Control (RBAC). The Sales Department head requested onboarding of new hire John Doe, requiring full integration into existing departmental workflow arrays and central update email streams.

ZENDESK TICKET

From: Sales Manager
Subject: Access Request: Add John Doe to Sales Groups
Body: Hi Help Desk, we have a new employee, John Doe, joining the team today. Please create his account and ensure he is added to the Sales-Group so he can access our files and receive team emails. Thanks!

Execution Strategy & Implementation

An Active Directory Security Group named Sales-Group was created in the Users container. A new corporate identity record was provisioned for John Doe (logon: jdoe) and nested directly within Sales-Group using ADUC's member management interface. The ticket was closed in Zendesk with a detailed resolution reply confirming full access provisioning.



Manager's onboarding request message.

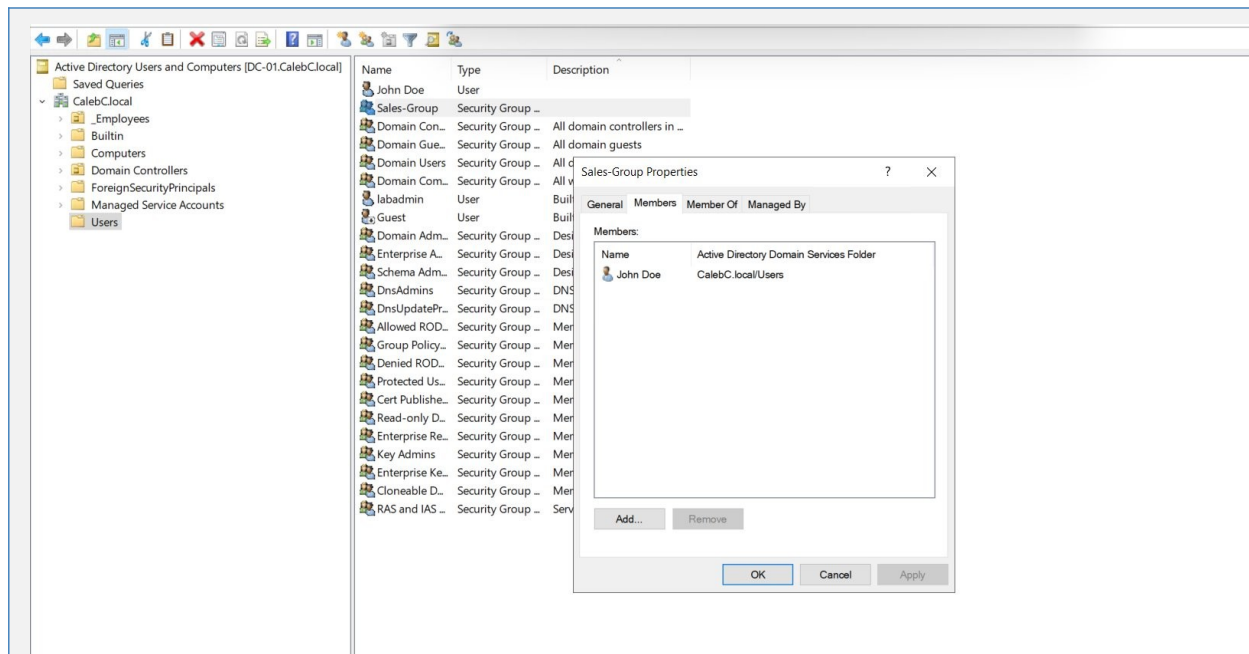


Figure: ADUC — Sales-Group Properties > Members tab showing John Doe (CalebC.local/Users) confirmed as a member of the Sales-Group security group on DC-01.CalebC.local.

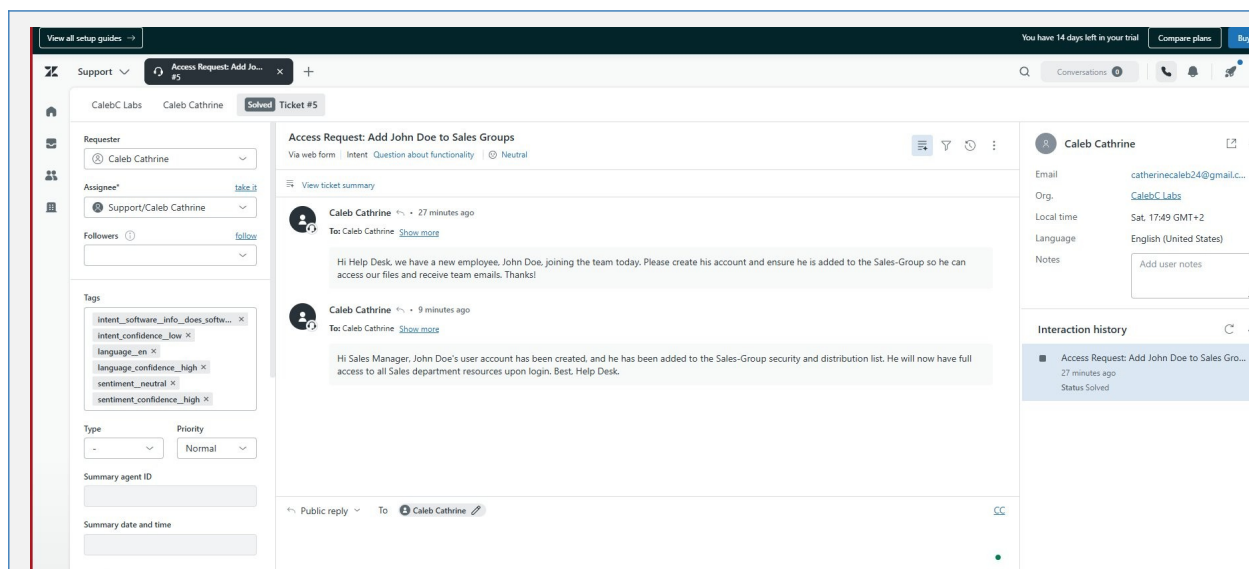


Figure: Zendesk Ticket #5 — marked Solved (Ticket #5). Full thread showing Sales Manager's request and Caleb Cathrine's resolution confirming John Doe's account creation and Sales-Group membership.

Technical Project Artifact Verification Log

The following table cross-references every screenshot captured during this lab with its corresponding infrastructure phase and the specific technical evidence it provides.

Image Filename	Category	Evidence Verified
Screenshot_2026-05-15_154307.png	Cloud Baseline	Azure VM Networking — DC-01-vnet, subnet 10.0.0.0/24, public IP, NIC NSG Basic confirmed.
Screenshot_2026-05-15_154615.png	VM Deployment	Azure Review + Create validation passed. Standard B2s v2 confirmed before provisioning.
Screenshot_2026-05-15_154938.png	Server Baseline	SConfig on DC-01 — computer name, RDP enabled, pre-promotion workgroup state confirmed.
Screenshot_2026-05-15_155601.png	DC Promotion	AD DS Config Wizard Prerequisites Check passed on DC-01 for CalebC.local.
Screenshot_2026-05-15_160738.png	Domain Active	ADUC — CalebC.local domain visible, DC promotion completed successfully.
Screenshot_2026-05-15_161957.png	Directory Layout	ADUC — _Employees OU with Jane Doe user object, OU structure confirmed.
Screenshot_2026-05-15_162425.png	Client Provisioning	Azure Marketplace — Windows 10 Enterprise image selection for Client-01.
Screenshot_2026-05-15_163021.png	Client Provisioning	Azure Marketplace — additional Windows VM image options reviewed for Client-01.
Screenshot_2026-05-15_163149.png	Domain Join	Computer Name/Domain Changes — CalebC.local joined, welcome popup confirmed.
Screenshot_2026-05-15_165714.png	Endpoint Validation	ADUC — Jane Doe confirmed in _Employees OU post domain join.
Screenshot_2026-05-16_154518.png	ITSM Setup	Zendesk — Jane Doe end-user profile created, Ticket #2 workspace ready.
Screenshot_2026-05-16_155147.png	File Share Config	CompanyShare Properties — network path \\DC-01\\CompanyShare active.
Screenshot_2026-05-16_155235.png	Drive Mapping	Client-01 — Z: drive (CompanyShare \\DC-01) mounted under Network Locations.
Screenshot_2026-05-16_163656.png	Ticket #2 Solved	Zendesk Ticket #2 Solved — drive mapping resolution confirmed in thread.
Screenshot_2026-05-16_165500.png	Ticket #3 Open	Zendesk Ticket #3 Urgent — Jane Doe lockout request received and assigned.
Screenshot_2026-05-16_165654.png	Password Reset	ADUC Reset Password — account unlocked, force change at next logon enabled.
Screenshot_2026-05-16_170150.png	Ticket #3 Solved	Zendesk Ticket #3 Solved — temp password Welcome2026! communicated.
Screenshot_2026-05-16_170441.png	Ticket #4 Open	Zendesk — HR offboarding ticket open, disable request received.
Screenshot_2026-05-16_170553.png	Account Disabled	ADUC — Jane Doe disabled icon confirms account de-provisioned.

Screenshot_2026-05-16_171539.png	Offboarding Closed	Zendesk — full lifecycle documented with all prior tickets in history.
Screenshot_2026-05-16_171714.png	Ticket #5 Open	Zendesk Ticket #5 — Sales Manager onboarding request for John Doe.
Screenshot_2026-05-16_173933.png	RBAC Membership	ADUC Sales-Group Members — John Doe confirmed nested in Sales-Group.
Screenshot_2026-05-16_174945.png	Ticket #5 Solved	Zendesk Ticket #5 Solved — John Doe account and Sales-Group membership confirmed.

✓ **VERIFICATION:** All 23 screenshots represent live, hands-on work performed in a real Azure cloud environment. No simulated or stock images were used. The domain CalebC.local, VM hostnames DC-01 and Client-01, and all user identities are genuine artefacts of this deployment.